SANGUINATE® (PP-007) – Highlighting the progress made on Haemoglobin based Oxygen carriers as a source of production of reliable artificial blood.

Gopal Agarwal, Hemant Jain, Isha Joshi, Ishaan Ghosh and Suryansh Sarangi

Abstract:

For centuries medical practitioners have tried to create a replacement for human blood, often using substances like plant resins, milk, beer, chicken or sheep blood. This topic has been prevalent for so long and still is a hot topic, with progress at a better rate over the past century. The aim was to replicate the oxygen carrying property of Haemoglobin in blood, thus the word Haemoglobin based oxygen carrier (HBOC). The modern alternative has reached the point where human trials have been addressed to and have stood from a positive standpoint. Thus we chose to represent SANGUINATE.

Introduction:

The search for a suitable blood substitute, after many attempts at artificial blood that failed to meet expectations has now led to Sanguinate. Developed by Prolong Inc., it is a unique PEG – modified form of bovine haemoglobin that represents the latest generation of HBOCs. It is able to endogenously deliver carbon monoxide, which has been shown to reduce inflammation and reperfusion injury also promote vasodialation.

PEG – Polyethylene glycol. The unique haemoglobin oxygen binding affinity of Sanguinate allows it to afford oxygen effective in the presence of hypoxide conditions.

Infraoperative nermavolemic haemodilution is a blood conservation technique that involves removal of whole blood from a patient with simultaneous replacement of the harvested blood with a fluid to maintain nermovolemia.

Method of Preparation:

SANGUINATE® (PP-007) is prepared through a multi-step bioprocess beginning with the isolation and purification of bovine hemoglobin from lysed red blood cells, which are clarified by centrifugation and microfiltration and further refined via ion-exchange and size-exclusion chromatography. The purified hemoglobin is then converted into carboxyhemoglobin by controlled exposure to low-pressure carbon monoxide gas, stabilizing the molecule and conferring its anti-inflammatory properties. Next, the carboxyhemoglobin is PEGylated through covalent linkage of NHS-activated polyethylene glycol (PEG) chains to accessible lysine residues on the globin, optimizing pharmacokinetics by extending circulatory half-life and minimizing immunogenicity. The resulting PEG–carboxyhemoglobin conjugate is buffer-exchanged into a physiologically compatible vehicle—balanced for pH, osmolality, and electrolyte composition—and supplemented with stabilizing antioxidants before being sterile-filtered (0.2 µm) and aseptically filled into single-use infusion vials under GMP conditions. Each lot undergoes comprehensive quality control, including spectrophotometric determination of carbon monoxide content, quantification of methemoglobin (< 1%), characterization of PEGylation profiles (via SEC-MALS or MALDI-TOF), endotoxin testing (< 0.5 EU/mL), and sterility and particulate analysis, ensuring that the final product meets stringent pharmaceutical-grade safety and efficacy standards.

1. **Isolation and Purification of Hemoglobin**
   - Lyse bovine red blood cells under cold, buffered conditions

- Clarify lysate by centrifugation and microfiltration
- Purify crude hemoglobin via sequential ion-exchange and size-exclusion chromatography

2. **Carboxyhemoglobin Formation**
   - Expose purified hemoglobin to low-pressure CO gas under controlled agitation
   - Stabilize as carboxyhemoglobin (HbCO)
   - Remove unbound CO by diafiltration
3. **PEGylation**
   - React HbCO with NHS-activated methoxy-PEG (5–10 kDa) targeting lysine residues
   - Control PEGylation to achieve ~2–4 PEG chains per hemoglobin tetramer
4. **Formulation and Sterile Filling**
   - Buffer-exchange PEG–CO–hemoglobin into physiologic formulation (balanced pH, osmolality, electrolytes)
   - Add antioxidants for protein stability
   - Sterile-filter final solution (0.2 µm)
   - Aseptically fill into single-use infusion vials under GMP
5. **Quality Control and Release Testing**
   - Verify carbon monoxide content via UV–Vis spectroscopy
   - Confirm methemoglobin percentage < 1%
   - Characterize PEGylation profile (SEC-MALS or MALDI-TOF)
   - Test endotoxin levels (< 0.5 EU/mL)
   - Perform sterility and particulate matter analysis

Chemical Properties:

1. Oxygen Delivery & Hemoglobin Mechanics

a) Oxygen Affinity (P_{50})

- **Definition**: The partial pressure of oxygen (pO_2) at which hemoglobin is 50% saturated.
- **Native Hemoglobin**: ~26 mmHg (cooperative binding due to tetrameric structure).
- **Sanguinate**: ~10–12 mmHg (**releases O_2 more easily in hypoxic tissues**).
- **Why It Matters**: Better for **stroke, trauma, or ischemia** where oxygen is critically low.

b) Carbon Monoxide (CO) Binding & Release

- **Native Hb**: CO binds irreversibly → toxic.
- **Sanguinate**:
  - **Controlled CO release → vasodilation & anti-inflammation**.
  - Helps in **sickle cell disease** and **organ transplants**.

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2. Pharmacokinetics (How the Body Processes It)

a) Half-Life & Circulation Time

Property	Sanguinate	Native Hb
Half-Life	20–24 hrs	<1 hr (free Hb)

Property	Sanguinate	Native Hb
Why?	PEGylation prevents rapid clearance.	

b) Clearance & Toxicity

- **Problem**: Free hemoglobin clogs kidneys (**nephrotoxic**).
- **Solution**: PEGylation prevents this → **safer excretion**.

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3. Safety & Side Effects

a) Vasoconstriction (Blood Vessel Narrowing)

- **Cause**: Free Hb scavenges nitric oxide (NO) → vessels tighten.
- **Sanguinate's Fix**: CO counters NO loss → **stable blood pressure**.

b) Oxidative Stress & Heme Toxicity

- **Heme Iron (Fe^{2+})** → Causes inflammation if free in blood.
- **PEG Shield** → Reduces heme reactivity → **less organ damage**.

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4. Clinical Advantages Over Blood Transfusions

Universal – No blood typing needed.
Long Shelf Life – >1 year at room temp vs. 42 days for blood.
No Infection Risk – Synthetic = no HIV/hepatitis.
Immediate Use – Critical for **trauma & emergencies**.

I. Molecular Design and Protein Engineering Principles

Sanguinate is constructed around **bovine carboxyhemoglobin (HbCO)**, deliberately stabilized through **site-directed PEGylation**. This involves covalent conjugation of **polyethylene glycol (PEG) chains** to surface lysine residues on the hemoglobin tetramer, fundamentally altering its biophysical behavior. The PEGylation strategy achieves three critical objectives:

1. **Steric Shielding**
   - PEG's hydrophilic polymer chains create a **hydration shell** around the hemoglobin molecule, reducing:
     - **Immunogenicity** (masking epitopes from immune recognition)
     - **Protein aggregation** (preventing dimerization/denaturation)
     - **Heme iron exposure** (limiting oxidative reactions)
   - The molecular weight increase (~90–200 kDa) pushes the complex beyond the **renal filtration threshold** (≥70 kDa), prolonging circulation.
2. **Controlled Carbon Monoxide (CO) Release**
   - The carboxyhemoglobin (HbCO) form is intentionally selected to exploit CO's **gasotransmitter properties**. CO dissociates slowly ($t_{1/2} \approx$ 20–24 hrs) via:
     - **Competitive displacement by O_2** in hypoxic tissues
     - **Photolytic cleavage** (minor pathway in vivo)

  - This yields **therapeutic CO concentrations** (0.1–1 μM), activating:
    - **sGC/cGMP pathway** → vasodilation
    - **HO-1 induction** → anti-inflammatory effects

3. **Heme Pocket Modulation**
  - PEGylation induces **conformational strain** on the globin chains, lowering O_2 affinity (P_{50} ≈ 10–12 mmHg vs. HbA's 26 mmHg). This is mechanistically distinct from:
    - **2,3-DPG modulation** (irrelevant in acellular Hb)
    - **Bohr effect attenuation** (PEG masks proton-sensitive residues)

II. Oxygen Transport Kinetics and Thermodynamics

Sanguinate operates under **non-cooperative O_2 binding kinetics**, diverging from the sigmoidal dissociation curves of native hemoglobin. This arises from:

- **Loss of quaternary structure constraints**: PEGylation "locks" hemoglobin in a relaxed (R) state, eliminating the T→R transition.
- **Fickian diffusion dominance**: O_2 release becomes linearly dependent on pO_2 gradients rather than allosteric triggers.

Clinical implications:

- Predictable O_2 unloading in **ischemic zones** (e.g., penumbra in stroke)

- No "overshoot" hypoxia risk from excessive O_2 extraction (a problem with high-affinity HBOCs)

III. Pharmacodynamic Interactions with Vascular Biology

Sanguinate’s hemodynamic neutrality stems from **dual gasotransmitter modulation**:

1. **Nitric Oxide (NO) Scavenging Mitigation**
    - Native cell-free Hb consumes NO via:

HbFe2+O2+NO→HbFe3++NO3−HbFe2+O2+NO→HbFe3++NO3−

    - PEG’s steric hindrance reduces NO access to heme by ~60% vs. unmodified Hb.
2. **CO-Mediated Compensation**
    - Released CO activates **soluble guanylate cyclase (sGC)** independently of NO:

sGC-Fe2++CO→sGC-Fe2+-CO→100×cGMP productionsGC-Fe2++CO→sGC-Fe2+-CO→100×cGMP production

    - Maintains basal vasodilation despite NO loss.

IV. Oxidative Stress and Heme Catabolism

The **PEG-heme iron interface** critically determines oxidative safety:

- **Fenton reaction suppression**:
  PEG’s ether oxygens coordinate Fe^{2+}, raising the activation barrier for:

H2O2+Fe2+→OH•+OH−+Fe3+H2O2+Fe2+→OH•+OH−+Fe3+

- **Haptoglobin-independent clearance**:
  PEG-HbCO bypasses the CD163-haptoglobin pathway, instead undergoing:
  - **RES phagocytosis** (slow, spleen-predominant)
  - **Renal elimination of PEG fragments** (MW <30 kDa)

V. Comparative Biophysics with Other Oxygen Carriers

Mechanism	Sanguinate	Perfluorocarbons	Polymerized Hb
O_2 Solubility	Hb-O_2 binding (1.34 mL O_2/g Hb)	Henry's law dissolution (~50 mL O_2/dL at 1 atm)	Hb-O_2 binding
pO_2 Dependency	Hyperbolic (non-cooperative)	Linear	Sigmoidal (cooperative)
Viscosity	2.5 cP (isooncotic)	0.8 cP (hemodilution risk)	3.2 cP (high shear)

Theoretical advantage: Sanguinate's **PEG-HbCO** combines gas carrier physiology with colloidal stability, avoiding the extreme pO_2 requirements of PFCs or the shear sensitivity of polymerized Hb.

VI. Unresolved Theoretical Questions

1. **Long-term PEG accumulation**: Potential lysosomal storage in RES macrophages.
2. **CO dosing precision**: Threshold between cytoprotection and mitochondrial inhibition.
3. **Shear-dependent O_2 release**: Whether flow turbulence modulates PEG-heme interactions.

Results of these tests:

To summarize, Sanguinate is a PEGylated bovine carboxyhemoglobin designed as a blood substitute with advanced oxygen delivery and therapeutic carbon monoxide (CO) release. Its low oxygen affinity (P_{50} ≈ 10–12 mmHg) allows efficient O_2 unloading in hypoxic tissues, unlike native hemoglobin. PEGylation extends its half-life (20–24 hrs), reduces kidney toxicity, and masks immunogenic sites. Sanguinate also offers vasodilation and anti-inflammatory effects via controlled CO release, benefiting conditions like sickle cell disease and ischemia. Unlike traditional transfusions, it is universally compatible, shelf-stable, and infection-free.

Molecularly, Sanguinate bypasses allosteric oxygen kinetics, ensuring predictable delivery under stress. PEG shielding diminishes oxidative damage and NO scavenging, while enabling non-cooperative O_2 diffusion. It avoids the high-pressure demands of perfluorocarbons and the viscosity issues of polymerized hemoglobin.

Research Work and Clinical Trials of SANGUINATE

Year	Phase/Study Type	Patient Population	Key Findings/Objectives
Pre-2013	Preclinical Animal Studies	N/A	Reduced myocardial infarct size, limited necrosis from cerebral ischemia, promoted recovery from hind limb ischemia.
2013	Phase I Clinical Trial	Healthy Volunteers	Established safety profile.
2014	Phase Ib Trial	Stable Sickle Cell Disease (SCD) Patients	Demonstrated acceptable safety profile.

2015	Phase Ib Trial	End-Stage Renal Disease (ESRD) Patients	Assessed safety, pharmacokinetics, and impact on humoral sensitisation.
2016	Phase II Trial	Sickle Cell Disease (Vaso-Occlusive Crisis) Patients	Ongoing trials for vaso-occlusive crisis treatment.
2017	Study	Patients with Subarachnoid Haemorrhage	Improved cerebral blood flow to vulnerable brain regions.
2017	Case Series	Acute Severe Anaemia Patients Without Blood Transfusion Options	Showed potential benefits.
2020	Immunological Study	ESRD Patients	Assessed safety, pharmacokinetics, and impact on humoral sensitisation.
2022	Study	Cystic Fibrosis Models	Demonstrated resolution of lung neutrophilia and decreased proinflammatory cytokines.

2023	Research	N/A	Enhanced collateral blood flow and anti-inflammatory effects.
2024-Ongoing	HEMERA-1 Phase I Clinical Trial	Acute Ischemic Stroke Patients	Evaluating safety and feasibility of Sanguinate (PP-007) in acute ischemic stroke.

- **Pre-2013** – *Preclinical Animal Studies*
  Preclinical trials demonstrated Sanguinate's efficacy in reducing myocardial infarct size, limiting necrosis from cerebral ischemia, and promoting recovery from hind limb ischemia.

- **2013** – *Phase I Clinical Trial in Healthy Volunteers*
  A Phase I trial involving healthy volunteers assessed the safety and pharmacokinetics of Sanguinate, establishing its safety profile.

- **2014** – *Phase Ib Trial in Stable Sickle Cell Disease (SCD) Patients*
  This trial evaluated the safety of Sanguinate in patients with sickle cell anemia, demonstrating an acceptable safety profile.

- **2015** – *Phase Ib Trial in End-Stage Renal Disease (ESRD) Patients*
  A study to assess the safety, pharmacokinetics, and

impact on humoral sensitization of Sanguinate infusion in ESRD patients.

- **2016** – *Phase II Trial in Sickle Cell Disease (Vaso-Occlusive Crisis)*
  Phase II trials for vaso-occlusive crisis treatment in SCD patients were ongoing and recruiting participants.

- **2017** – *Cerebral Blood Flow Study After Subarachnoid Hemorrhage*
  A study demonstrated that Sanguinate improved cerebral blood flow to vulnerable brain regions at risk of delayed cerebral ischemia after subarachnoid hemorrhage.

- **2017** – *Case Series: Acute Severe Anemia Patients Without Blood Transfusion Options*
  Sanguinate was used in patients with acute severe anemia who could not receive transfusions, showing potential benefits.

- **2020** – *Immunological Study in ESRD Patients*
  A study assessed the safety, pharmacokinetics, and impact on humoral sensitization of Sanguinate infusion in patients with end-stage renal disease.

- **2022** – *Pulmonary Inflammation Study in Cystic Fibrosis Models*

A study demonstrated that Sanguinate induces heme oxygenase-1 (HO-1) expression, leading to the resolution of lung neutrophilia and decreased proinflammatory cytokines in cystic fibrosis models.

- **2023** – *Therapeutic Induction of Collateral Flow*
  Research indicated that Sanguinate causes vasodilation of leptomeningeal arteries and releases small amounts of carbon monoxide, enhancing collateral blood flow and providing anti-inflammatory effects.

- **2024–Ongoing** – *HEMERA-1 Phase I Clinical Trial in Acute Ischemic Stroke Patients*
  HEMERA-1 is a Phase I clinical trial evaluating the safety and feasibility of Sanguinate (PP-007) in patients with acute ischemic stroke.

Precautionary findings in these Trial:

Area of Concern	**Studies/Observations**	**Findings/Implications**
Cardiac Marker Elevations	Phase Ib in Sickle Cell Disease (2014)	Some patients exhibited transient elevations in troponin levels, a cardiac marker. While these didn't always correlate with overt cardiac events, they raised safety concerns.

Cardiac Marker Elevations	Phase Ib in End-Stage Renal Disease (2015)	Similar troponin elevations were observed. Few experienced a non-ST elevation myocardial infarction (NSTEMI), though causality with Sanguinate remained unclear.

Future Prospects:

1. Treatment of Sickle Cell Disease (SCD)
   - FDA orphan drug status for SCD.
   - Shown to improve oxygenation, reduce vaso-occlusion, and lower pain during crises.
   - Could reduce hospitalizations and improve quality of life.
   - *Clinical trials ongoing for pain crises and anemia in SCD patients.*
2. Ischemia-Reperfusion Injuries
   - Heart attacks, strokes, organ transplants.
   - Protects tissues during low-oxygen conditions and controlled reoxygenation.
   - Potential use during organ preservation for transplantation.
3. Trauma and Emergency Care
   - Rapid oxygen delivery when blood transfusion isn't possible.
   - Battlefield medicine, rural settings, mass casualty incidents.

- Alternative to donated blood—longer shelf life, no need for cross-matching.

4. Cancer and Tumor Hypoxia
   - Some early research explores HBOCs in overcoming hypoxia in tumors, improving the effect of radiation or chemotherapy.
   - Not yet a primary focus, but potential niche.
5. Renal and Cardiovascular Protection
   - Studies suggest PEGylated CO-Hb helps prevent acute kidney injury and vascular damage.
   - May aid in chronic kidney disease or diabetic vasculopathy management in the future.

Challenges & Limitations

- Regulatory Approval: HBOCs have historically struggled due to toxicity concerns (e.g., vasoconstriction, oxidative stress).
- Cost of production: PEGylation and purification are complex.
- Competition from other oxygen therapeutics: e.g., Hemopure, Oxyglobin, perfluorocarbon-based carriers.

Market Value:

Since a product needs to hold a significant potential future in terms of its market value apart from its benefits as a product and in general well-being towards society. A peek at the potential market value must be analyzed not just for Sanguinate, but the entire research involving artificial blood.

Synthetic Red Blood Cells (HBOCs) Market

The global synthetic red blood cell (RBC) market, driven by Hemoglobin-Based Oxygen Carriers (HBOCs) like Sanguinate, was valued at $1.2 billion in 2024 and is projected to reach $2.5 billion by 2030, growing at a CAGR of 11.5%. North America leads with 45% market share ($540M), followed by Europe (30%) and Asia-Pacific (fastest-growing, 13.2% CAGR).

Key players include Prolong Pharmaceuticals (35% market share, $420M revenue), Northfield Laboratories (25%, $300M), and Hemopure (20%, $240M). Synthetic RBCs significantly cut costs by reducing $500M in logistics, minimizing $300M wastage, and eliminating $200M in infection-related expenses.

The market is fueled by global blood shortages (meeting only 70% of demand), R&D investments ($600M in 2024), and extended shelf life (1-year vs. 42 days for real blood). Production costs are declining, with HBOC unit prices expected to drop from $1,200 (2024) to $800 (2030). Investors are seeing 15-20% ROI, with nanoparticle RBCs growing at 18.7% CAGR. Asia-Pacific presents the highest growth opportunity, while government incentives and hospital adoption drive expansion.

Conclusion:

Sanguinate represents a promising synthetic oxygen carrier with unique pharmacologic and biophysical benefits. While offering clinical advantages in emergency and ischemic scenarios, long-term safety and CO dosing precision remain critical areas for future research.

Sanguinate (PP-007), a PEGylated carboxyhemoglobin, has undergone extensive research and clinical evaluation for various ischemic and inflammatory conditions. Preclinical studies prior to 2013 showed its potential in reducing tissue damage in myocardial infarction, cerebral ischemia, and limb ischemia. Clinical trials began in 2013, confirming its safety in healthy volunteers, sickle cell disease (SCD), and end-stage renal disease (ESRD) patients. Further studies explored its application in vaso-occlusive crises, subarachnoid hemorrhage, acute anemia, and cystic fibrosis, where it demonstrated improved oxygen delivery, anti-inflammatory effects, and cerebral or collateral blood flow enhancement.

However, some concerns were raised. Cardiac marker elevations, including troponin, were observed in certain trials, with one NSTEMI case reported in an ESRD patient. Additionally, short-term cerebral perfusion improvements were not sustained beyond 24 hours. Immunological monitoring is advised due to potential humoral sensitization risks.

Sanguinate shows promise as a therapeutic alternative in critical care, particularly for patients who cannot receive blood transfusions or suffer from ischemic conditions. Its dual mechanism—oxygen delivery and CO-mediated vasodilation—offers broad clinical utility. While the safety profile is generally favorable, careful dosing, cardiac monitoring, and long-term immunologic evaluation are essential to optimize its clinical use and address lingering safety concerns.

Contribution:

Abstract, Future prospects, compilation and editing (Synthesizer): Ishaan Ghosh

Method of preparation and idea: Isha Joshi

Chemical Properties: Hemant Jain

Data on clinical trials and market value: Suryansh Sarangi

Introduction and Conclusion: Gopal Agarwal

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